MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (ACCURATE)

ESTABLISHED RELIABILITY

STYLES RBR80 AND RBR81

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RBR80 and RBR81, established reliability, accurate, wire-wound, fixed resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-39005 - Resistors, Fixed, Wire-Wound (Accurate), Established Reliability, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-39005, and as specified herein.

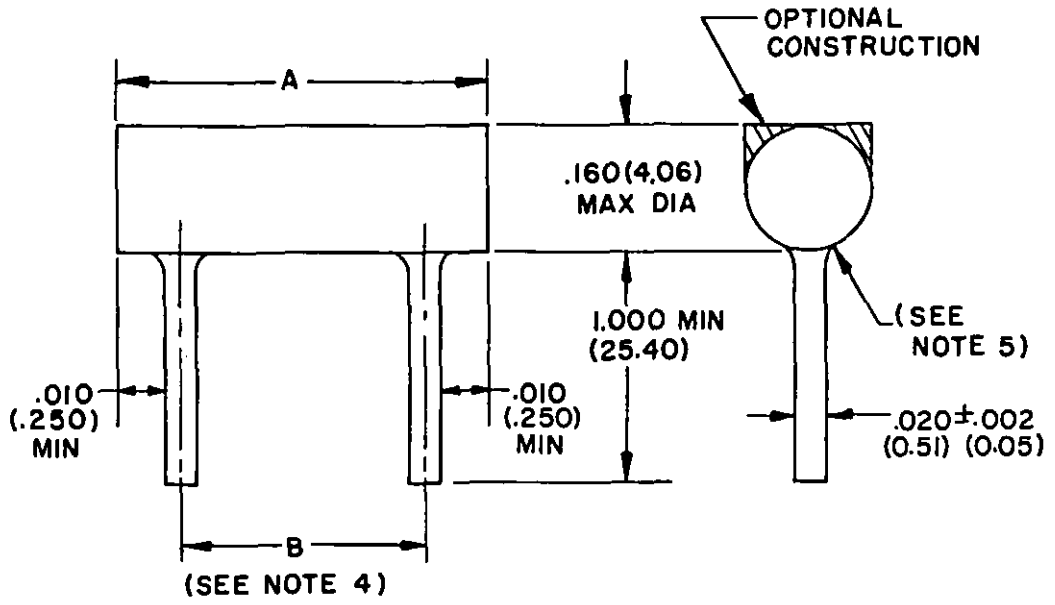
3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Minimum resistance value and applicable tolerance. Minimum resistance values and applicable tolerances are as follows:

Minimum resistance (ohms) 1/	Resistance tolerance (percent)
1000	T (±.01)
100	Q (±.02) and A (±.05)
10	B (±.10) and F (±1.0)

1/ Minimum resistance values are available for both styles RBR80 and RBR81, but it is preferred that style RBR81 be limited in use to resistance values exceeding .120 megohm.

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Style	A, Max	B ±.005 (0.13)
RBR80	.325 (8.26)	.225 (5.72)
RBR81	.500 (12.70)	.400 (10.16)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parenthesis.
4. The lead measurements shall be made at the point of emergence from the body.
5. Leads are to be clean and solderable to within .005 (0.13 mm) from the resistor body.

FIGURE 1. Styles RBR80 and RBR81 resistors.

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3.4 Maximum resistance value. Maximum resistance values shall be as follows:

Resistor style	Maximum resistance 1/ (megohms)
RBR80	.120
RBR81	.250

1/ Based on use of 0.0006-inch or larger diameter wire.

3.5 Power rating. Power rating shall be 1/10 watt for styles RBR80 and RBR81.

3.6 Maximum voltage. Maximum voltage shall be 100 volts direct current or peak for styles RBR80 and RBR81.

* 3.7 Maximum weight. The maximum weight shall be .4 gram for style RBR80 and .55 gram for style RBR81.

* 3.8 Resistance-temperature characteristic. The resistance-temperature characteristic shall not exceed the value listed in table I.

TABLE I. Maximum resistance-temperature characteristic.

Resistance (ohms)	Parts/million/degree C (PPM/°C
Less than 100 ohms	±10
100 ohms and above	± 5

* 3.9 Maximum allowable reactance (see 4.3). The maximum allowable reactance shall be defined as follows:

(a) For resistance values less than 500 ohms, the phase angle shall not exceed +5 degrees.
(b) For resistance values 500 ohms or greater, but less than 20,000 ohms, the phase angle shall not exceed ±10 degrees.
(c) For resistance values 20,000 ohms or greater, but less than 50,000 ohms, the effective parallel capacitance shall not exceed 4.0 pF.
(d) For resistance values of 50,000 ohms or greater, the maximum effective parallel capacitance shall not exceed 6.0 pF.

* 3.10 Life. The maximum allowable change in resistance after 2,000 hours (qualification inspection) shall be as follows:

(a) For resistance values of 10 ohms or greater, but less than 100 ohms, the change in resistance shall not exceed ±0.1 percent.
(b) For resistance values of 100 ohms or greater, the change in resistance shall not exceed ±0.05 percent.

* 3.11 High temperature exposure. The maximum allowable change in resistance after 2,000 hours exposure to an ambient of 145°C shall be as follows:

(a) For resistance values of 10 ohms or greater, but less than 100 ohms, the change in resistance shall not exceed ±0.1 percent.
(b) For resistance values of 100 ohms or greater, the change in resistance shall not exceed ±0.05 percent.

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* 3.12 Marking. Due to size limitations, style RBR80 resistor shall be marked with the following minimum information:

BR80L	-	Partial style and terminal designator
12701	-	Coded resistance value
FMJ	-	Tolerance, failure rate, JAN marking
7445A	-	Date code, lot code

The complete marking is required on the unit package. Where manufacturers are able to provide more information, the following is preferred in the sequence presented: Style and source code.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Qualification inspection. Qualification inspection shall be in accordance with MIL-R-39005. In addition, following the completion of group V of table VII, the 30 samples shall be tested as specified in 4.3, herein, with zero defectives allowed.

* 4.2 Quality conformance inspection. Sampling and inspection shall be in accordance with MIL-R-39005. In addition, prior to subjecting the 102 sample units to the annual group C (table X) "High temperature exposure" test, 30 of these samples (10-high value, 10-10,000 ohms, 10-low value) will be tested as specified in 4.3, herein, with zero defectives allowed.

4.3 Maximum allowable reactance. The inherent reactance shall be measured as follows:

* (a) Test instrument - Hewlett-Packard RX meter, model 250B; Tektronic LC meter, type 130, or any other equivalent meter.
(b) Frequency range - 100 kHz to 500 kHz.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-39005.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-39005 are applicable to this specification.

6.2 Supplementary insulation. Where the potential to ground is over 250 volts, the supplementary insulation should be provided.

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5905-N262)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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